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LIGHT EMITTING CHIP INCLUDING A PLURALITY OF LIGHT EMITTING UNITS AND IMAGE FORMING APPARATUS

Abstract

A light emitting chip includes: a first data storage unit configured to store data for causing a first light emitting unit to emit light; a second data storage unit configured to store data for causing a second light emitting unit to emit light; a signal output unit configured to output a first trigger signal and a second trigger signal delayed with respect to the first trigger signal; and a drive unit configured to drive the first light emitting unit, in a case where the first trigger signal is output, based on data stored in the first data storage unit, and drive the second light emitting unit, in a case where the second trigger signal is output, based on data stored in the second data storage unit.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a light emitting chip including a plurality of light emitting units, and an image forming apparatus that forms an image using the light emitting chip.

Description of the Related Art

[0002] An electrophotographic image forming apparatus forms an electrostatic latent image on the photoconductor by exposing a rotationally driven photoconductor, and forms an image by developing the electrostatic latent image with toner. Here, a direction parallel to the rotation axis of the photoconductor is referred to as a main scan direction. Japanese Patent Laid-Open No. 2021-035765 discloses an image forming apparatus that performs exposure by using an exposure apparatus including a plurality of columns of light emitting units arranged in a sub scan direction corresponding to a rotation direction of a photoconductor, in which each column includes a plurality of light emitting units arranged in a main scan direction. In Japanese Patent Laid-Open No. 2021-035765, flip-flop circuits configured to hold image data are respectively provided in corresponding number as number of light emitting units, and when image data is held in all the flip-flop circuits, control of the light emitting units are performed all at once. Accordingly, there is a concern in the aforementioned exposure head that simultaneously turning on a large number of light emitting units may cause electrical noise such as switching noise to increase.

SUMMARY OF THE INVENTION

[0003] The present disclosure reduces occurrence of electrical noise due to simultaneously turning on a plurality of light emitting units in an image forming apparatus including an exposure head provided with a plurality of light emitting units.

[0004] According to an aspect of the present invention, a light emitting chip includes: a light emitting unit array including a plurality of light emitting units arranged in a predetermined direction; an image data input terminal configured to be input an image data; a first data storage unit configured to store data, input from the image data input terminal, for causing a first light emitting unit included in the plurality of light emitting units to emit light; a second data storage unit configured to store data, input from the image data input terminal, for causing a second light emitting unit included in the plurality of light emitting units to emit light; a signal output unit configured to output a first trigger signal and a second trigger signal delayed with respect to the first trigger signal; and a drive unit configured to drive the first light emitting unit, in a case where the first trigger signal is output, based on data stored in the first data storage unit, and drive the second light emitting unit, in a case where the second trigger signal is output, based on data stored in the second data storage unit.

[0005] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a schematic configuration diagram of an image forming apparatus according to an embodiment;

[0007] FIGS. **2**A and **2**B are diagrams illustrating an exposure head and a photoconductor according to an embodiment;
[0008] FIGS. **3**A and **3**B are diagrams illustrating a printed substrate of an exposure head according to an embodiment;
[0009] FIG. **4** is an explanatory diagram of an arrangement of light emitting elements in a light emitting chip, according to an embodiment;
[0010] FIG. **5** is a plan view of a light emitting chip, according to an embodiment;
[0011] FIG. **6** is a cross sectional view of a light emitting chip, according to an embodiment;
[0012] FIG. **7** is a control configuration diagram of a light emitting chip, according to an embodiment;
[0013] FIG. **8** is diagram illustrating example signals on respective signal lines when a register of a light emitting chip is accessed, according to an embodiment;
[0014] FIG. **9** is diagram illustrating example signals on respective signal lines when image data is transmitted to a light emitting chip, according to an embodiment;
[0015] FIG. **10** is a functional block diagram of a light emitting chip, according to an embodiment;
[0016] FIG. **11** is a configuration diagram of a transfer unit, according to an embodiment;
[0017] FIG. **12** is a configuration diagram of a latch unit, according to an embodiment;
[0018] FIG. **13** is a configuration diagram of another latch unit, according to an embodiment;
[0019] FIG. **14** is a timing chart of signals in a circuit unit, according to an embodiment;
[0020] FIG. **15** is a configuration diagram of a current drive unit according to an embodiment;
[0021] FIG. **16** is a flowchart of processing executed by an image controller, according to an embodiment; and
[0022] FIG. **17** is another timing chart of signals in a circuit unit according to an embodiment.

DESCRIPTION OF THE EMBODIMENTS

[0023] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate.
[0024] Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

[0025] FIG. **1** is a schematic configuration diagram of an image forming apparatus according to the present embodiment. A reading unit **100** reads a document placed on a document plate and generates image data representing a reading result. An image creating unit **103** forms an image on a sheet based on, for example, the image data generated by the reading unit **100** or image data received from an external apparatus via a network.
[0026] The image creating unit **103** includes image forming units **101***a*, **101***b*, **101***c* and **101***d*. The image forming units **101***a*, **101***b*, **101***c* and **101***d* respectively form black, yellow, magenta and cyan toner images. The image forming units **101***a*, **101***b*, **101***c* and **101***d* have a similar configuration and therefore will be collectively referred to as an image forming unit **101**. In forming an image, a photoconductor **102** of the image forming unit **101** is rotationally driven in a clockwise direction of the diagram. A charger **107** charges the photoconductor **102**. An exposure head **106**, which is an exposure apparatus, exposes the photoconductor **102** in accordance with the image data, and forms an electrostatic latent image on the photoconductor **102**. A developer **108** develops the electrostatic latent image on the photoconductor **102** with toner. The toner image formed on the photoconductor **102** is transferred onto a sheet that is conveyed on a transfer belt **111**. Here, different colors from black, yellow, magenta and cyan can be reproduced by transferring toner images formed on respective photoconductors **102** onto a sheet in an overlapping manner.
[0027] A conveyance unit **105** controls feed and conveyance of a sheet. Specifically, the

conveyance unit **105** feeds a sheet to the conveyance path of the image forming apparatus from a unit specified among internal storage units **109***a* and **109***b*, an external storage unit **109***c*, and a manual feed unit **109***d*. The sheet being fed is conveyed to a registration roller **110**. The registration roller **110** conveys the sheet onto the transfer belt **111** at a predetermined timing such that toner images formed on respective photoconductors **102** are transferred onto the sheet. As has been described above, the toner images are transferred onto the sheet while the sheet is being conveyed on the transfer belt **111**. A fixing unit **104** heats and pressurizes the sheet, on which the toner images are transferred, to fix the toner images on the sheet. After the toner images are fixed, the sheet is discharged by a discharge roller **112** to the outside of the image forming apparatus.

[0028] FIGS. **2**A and **2**B illustrate the photoconductor **102** and the exposure head **106**. The exposure head **106** includes a light emitting point group **201**, a printed substrate **202** in which the light emitting point group **201** is mounted, a rod lens array **203**, and a housing **204** that supports the rod lens array **203** and the printed substrate **202**. The rod lens array **203** focuses the light emitted from the light emitting point group **201** onto the photoconductor **102** to form an imaging spot of a predetermined size on the photoconductor **102**.

[0029] FIGS. **3**A and **3**B illustrate the printed substrate **202**. Here, FIG. **3**A illustrates a surface on which a connector **305** is mounted, and FIG. **3**B illustrates a surface on which the light emitting point group **201** is mounted (an opposite surface to the surface on which the connector **305** is mounted). In the present embodiment, the light emitting point group **201** includes 20 light emitting chips **400-1** to **400-20**. The light emitting chips **400-1** to **400-20** are arranged in two staggered columns along the main scan direction. In the following description, the light emitting chips **400-1** to **400-20** will also be collectively referred to as light emitting chips **400**. Each of the light emitting chips **400** includes a plurality of light emitting points (light emitting elements). Each of the light emitting chips **400** of the printed substrate **202** is connected, via a connector **305**, to an image controller **700** (FIG. **7**), which is a control unit.

[0030] FIG. **4** is an explanatory diagram of an arrangement of the light emitting chips **400** and light emitting points **602** provided on the light emitting chips **400**. One light emitting chip **400** includes a plurality of sets of 748 light emitting points **602** arranged along the main scan direction. Here, the plurality of sets are arranged along a sub scan direction orthogonal to the main scan direction. As such, the light emitting chips **400** are arranged two-dimensionally along both the main scan direction and the sub scan direction. In other words, the light emitting points **602** may be arranged by including M rows in the circumferential direction of the photoconductor and N columns in the axial direction of the photoconductor. In the following description, the number of sets is assumed to be four, as an example. In other words, it is assumed in the following exemplary embodiment that the light emitting chips **400** include four sets of 748 light emitting points **602** arranged along the main scan direction, i.e., a total of 2992 light emitting points **602**. The pitch of the light emitting points **602** adjacent to each other in the main scan direction is about 21.16 μm, which corresponds to a resolution of 1200 dpi. Therefore, the length of one set of 748 light emitting points **602** in the main scan direction is about 15.8 mm. In addition, the pitch (length P in FIG. **4**) of the light emitting points **602** adjacent to each other in the sub scan direction is about 21.16 μm, which corresponds to a resolution of 1200 dpi. Furthermore, the pitch (length L in FIG. **4**) between the light emitting points **602** of two light emitting chips **400** adjacent to each other in the main scan direction is about 21.16 μm, which corresponds to a resolution of 1200 dpi.

[0031] FIG. **5** is a plan view of the light emitting chip **400**. The plurality of light emitting points **602** of the light emitting chip **400** are formed, for example, on a light emitting substrate **402**, which is a silicon substrate. Furthermore, a circuit unit **406** configured to control the plurality of light emitting points **602** is formed on the light emitting substrate **402**. A signal line configured to communicate with the image controller **700**, a power line configured to connect to a power source, and a ground line configured to connect to the ground will be connected to pads **408-1** to **408-9**. The signal line, the power line, and the ground line are wires made of gold, for example.

[0032] FIG. **6** illustrates a part of a cross section taken along line A-A in FIG. **5**. A plurality of lower electrodes **504** are formed on the light emitting substrate **402**. A gap of a length d is provided between two adjacent lower electrodes **504**. A light emitting layer **506** is provided on the lower electrodes **504**, and an upper electrode **508** is provided on the light emitting layer **506**. The upper electrode **508** is one common electrode for the plurality of lower electrodes **504**. When a predetermined voltage is applied between the lower electrode **504** and the upper electrode **508**, electric current flows from the lower electrode **504** to the upper electrode **508**, and thus the light emitting layer **506** emits light. Therefore, a region of the light emitting layer **506** corresponding to a region of one lower electrode **504** corresponds to one light emitting point **602**. In other words, the light emitting substrate **402** of the present embodiment includes a plurality of light emitting points. Here, the light emitting point may also be referred to as a light emitting unit. A light emitting unit array including a plurality of light emitting units arranged in the main scan direction is formed in each of the light emitting chips **400**.
[0033] Organic EL film, for example, may be used for the light emitting layer **506**. Alternatively, an inorganic EL film may be used for the light emitting layer **506**. The upper electrode **508** is formed of, for example, a transparent electrode made of indium tin oxide (ITO) or the like, for transmitting the emission wavelength of the light emitting layer **506**. Although the whole of the upper electrode **508** transmits the emission wavelength of the light emitting layer **506** in the present embodiment, the whole of the upper electrode **508** needs not transmit the emission wavelength. Specifically, a part of the upper electrode **508** corresponding to a region where light is emitted from each light emitting point **602** may transmit the light emission wavelength.
[0034] Although one continuous light emitting layer **506** is formed in FIG. **6**, a plurality of light emitting layers **506** each having the similar width as the width W of the lower electrode **504** may be respectively formed on the lower electrode **504**. In addition, although the upper electrode **508** is one common electrode for the plurality of lower electrodes **504** in FIG. **6**, a plurality of upper electrodes **508** each having the similar width as the width W of the lower electrode may be formed as corresponding to each of the lower electrodes **504**. Alternatively, a first plurality of lower electrodes **504** among the lower electrodes **504** of each of the light emitting chips **400** may be covered by a first light emitting layer **506**, and a second plurality of lower electrodes **504** may be covered by a second light emitting layer **506**. Alternatively, a first upper electrode **508** may be commonly formed as corresponding to a first plurality of lower electrodes **504** and a second upper electrode **508** may be commonly formed as corresponding to a second plurality of lower electrodes **504**, among the lower electrodes **504** of each of the light emitting chips **400**. Also in the aforementioned configuration, one lower electrode **504**, a region of the light emitting layer **506** and the upper electrode **508** corresponding to the one lower electrode **504** form one light emitting point (light emitting element) **602**. Here, the light emitting point **602** may also be referred to as a light emitting unit.
[0035] FIG. **7** illustrates a control configuration of the light emitting chips **400**. A data switching unit **705** and each of the light emitting chips **400** are connected by a plurality of signal lines (wires). Specifically, the data switching unit **705** and the light emitting chip **400**-*n* (n being an integer from 1 to 20) are connected by a signal line DATAn and a signal line WRITEn. The signal line DATAn is used by the data switching unit **705** to transmit image data to the light emitting chip **400**-*n*. The signal line WRITEn is used by the data switching unit **705** to write control data to a register of the light emitting chip **400**-*n*.
[0036] In addition, the data switching unit **705** and all the light emitting chips **400** are connected by one signal line CLK, one signal line SYNC, and one signal line EN. The signal line CLK is used for transmitting a clock signal for data transmission in the signal lines DATAn and WRITEn. The data switching unit **705** outputs, to the signal line CLK, a clock signal generated based on a reference clock signal from a clock generator **702**. Signals transmitted to the signal line SYNC and the signal line EN will be described below.

[0037] A CPU **701** controls the image forming apparatus **100** as a whole. An image data generation unit **703** performs various types of image processing such as halftone processing on the image data received from the reading unit **100** or an external apparatus, and generates image data for performing ON/OFF control of light emission from the light emitting point **602** of each of the light emitting chips **400**. The image data generation unit **703** transmits the generated image data to the data switching unit **705**. A register access unit **704** receives, from the CPU **701**, control data to be written to the register of each of the light emitting chips **400**, and transmits the control data to the data switching unit **705**.
[0038] FIG. **8** illustrates signals of the respective signal lines when control data is written to registers of the light emitting chips **400**. An enable signal, which rises to a high level indicating in-communication during communication, is output to the signal line EN. The data switching unit **705** transmits a start bit to the signal line WRITEn in synchronization with the rising of the enable signal.
[0039] Subsequently, the data switching unit **705** transmits a write identification bit indicating write operation, and subsequently transmits an address (4 bits in the present example) of the register, to which control data is written, and control data (8 bits in the present example). In writing to the register, the data switching unit **705** set the frequency of the clock signal transmitted to the signal line CLK to be 3 MHz, for example.
[0040] FIG. **9** illustrates signals of respective signal lines when image data is transmitted to each of the light emitting chips **400**. A cyclic line synchronization signal indicating an exposure timing of one line in the photoconductor **102** is output to the signal line SYNC. Assuming that the circumferential speed of the photoconductor **102** is 200 mm/s and the resolution in the sub scan direction is 1200 dpi (about 21.16 μm), the line synchronization signal is output at a cycle of about 105.8 μs. The data switching unit **705** transmits the image data to signal lines DATA**1** to DATA**20** in synchronization with the rising of the line synchronization signal. In the present embodiment, since each of the light emitting chips **400** includes 2992 light emitting points **602**, it is necessary to transmit image data, to each of the light emitting chips **400**, indicating emission/non-emission of light for each of a total of 2992 light emitting points **602** within a cycle of about 105.8 μs. In order to transmit image data for a total of 2992 light emitting points **602** within a period of about 105.8 μs, the data switching unit **705** sets, in transmitting image data, the frequency of the clock signals transmitted to the signal line CLK to 30 MHz in the present example, as illustrated in FIG. **9**.
[0041] FIG. **10** is a functional block diagram of one light emitting chip **400**-*n*. As also illustrated in FIG. **5**, the light emitting chip **400** includes nine pads **408-1** to **408-9**. The pads **408-1** and **408-2** are connected to a power supply voltage VCC by a power line. Electric power is supplied to each circuit of the circuit unit **406** of the light emitting chips **400** by the power supply voltage VCC. The pads **408-3** and **408-4** are connected to the ground by a ground line. Each circuit of the circuit unit **406** and the upper electrode **508** are connected to the ground via the pads **408-3** and **408-4**. The signal line CLK is connected to a transfer unit **1003**, a register **1102**, and latch units **1004-001** to **1004-748** via the pad **408-5**. The signal lines SYNC and DATAn are respectively connected to the transfer unit **1003** via the pads **408-6** and **408-7**. In other words, the pad **408-6** corresponds to a synchronization signal input terminal, and the pad **408-7** corresponds to an image data input terminal. The signal lines EN and WRITEn are respectively connected to the register **1102** via the pads **408-8** and **408-9**. As has been described above, the register **1102** stores control data indicating control information. Details of the control information will be described below.
[0042] In the following, the operation of the transfer unit **1003** and the latch units **1004-001** to **1004-748** (also collectively referred to as latch units **1004**) will be described, referring to FIGS. **11** to **14**. Here, it is assumed that the transfer unit **1003** receives, with the line synchronization signal from the signal line SYNC as a starting point and in synchronization with the clock signal, pieces of image data each indicating emission/non-emission of light of one light emitting point **602** in the order of D**1**, D**2**, D**3**, D**2992**. FIG. **14** illustrates a state in which the transfer unit **1003** has received

pieces of image data D**1** to D**12**.

[0043] FIG. **11** is a configuration diagram of the transfer unit **1003**. A clock signal from the signal line CLK, a line synchronization signal from the signal line SYNC, and image data from the signal lines DATAn are input to the transfer unit **1003**. D flip-flops **1101-1** to **1101-8** operate in accordance with the clock signal. The D flip-flop **1101-1** sequentially outputs the image data received from DATAn to a signal line PDATA**4**. The D flip-flop **1101-2** outputs the image data received from DATAn to a signal line PDATA**3** by delaying one clock from the image data output from the D flip-flop **1101-1**. Similarly, the D flip-flops **1101-3** and **1101-4** output the image data received from DATAn to signal lines PDATA**2** and PDATA**1** by respectively delaying two clocks and three clocks from the image data output from the D flip-flop **1101-1**.

[0044] FIG. **14** illustrates image data output from the transfer unit **1003** to the signal lines PDATA**1** to PDATA**4**. As illustrated in FIG. **14**, the transfer unit **1003** outputs the pieces of image data D**3**, D**2** and D**1** respectively to the signal lines PDATA**3**, PDATA**2** and PDATA**1** at the timing when the transfer unit **1003** outputs the piece of image data D**4** to the signal line PDATA**4**. More generally, the transfer unit **1003** outputs the pieces of image data D(4m−1), D(4m−2) and D(4m−3) respectively to the signal lines PDATA**3**, PDATA**2** and PDATA**1** at the timing when the transfer unit **1003** outputs the piece of image data D(4m) to the signal line PDATA**4**, where m being an integer from 1 to 748.

[0045] The transfer unit **1003** delays the line synchronization signal by the D flip-flops **1101-5** to **1101-8**, and outputs a first latch signal to a signal line LAT**1** at a timing when the transfer unit **1003** outputs D**4** to the signal line PDATA**4**.

[0046] FIG. **12** is a configuration diagram of the latch unit **1004-001**. D flip-flops **1202-1** to **1202-4** and latch circuits **1201-1** to **1201-4** operate in accordance with the clock signal from the signal line CLK. The image data from the signal lines PDATA**4** to PDATA**1** and the first latch signal from the signal line LAT**1** are input to the latch circuits **1201-1** to **1201-4**. Upon receiving the first latch signal from the signal line LAT**1**, the latch circuits **1201-1** to **1201-4** latch the values being output to the signal lines PDATA**4** to PDATA**1** at that time, and output drive signals based on the latched values to the signal lines PON**1-4** to PON**1-1**, as illustrated in FIG. **14**. That is, the latch circuits **1201-1** to **1201-4** store the input image data, and output drive signals based on the stored image data. In other words, the latch circuits **1201-1** to **1201-4** correspond to a first data storage unit or a plurality of data storage units. Furthermore, the first latch signal corresponds to a first trigger signal. According to FIG. **14**, the signal lines PON**1-1** to PON**1-4** output drive signals based on the pieces of image data D**1** to D**4**. In the present embodiment, one light emitting chip **400** includes the structure in which four sets of 748 light emitting points **602** arranged along the main scan direction are arranged along the sub scan direction, as has been described referring to FIG. **4**. For example, the light emitting points **602** of each set are numbered 1 to 748 along the X direction in FIG. **5** (corresponding to the main scan direction). In this case, the respective drive signals output to the signal lines PON**1-4** to PON**1-1** are drive signals of each first light emitting point **602** of the four sets. The drive signals are input to the current drive unit **1104**. The latch unit **1004-001** maintains outputting drive signals based on the pieces of image data D**1** to D**4** until receiving a first latch signal based on the next line synchronization signal. In other words, the drive signals based on the pieces of image data D**1** to D**4** are output for the duration of two consecutive line synchronization signals (about 105.8 us in the present example). In addition, the latch unit **1004-001** delays the first latch signal by the D flip-flops **1202-1** to **1202-4**, and outputs a second latch signal to a signal line LAT**2** at a timing when the transfer unit **1003** outputs D**8** to the signal line PDATA**4**.

[0047] FIG. **13** is a configuration diagram of the latch unit **1004-002**. D flip-flops **1302-1** to **1302-4** and latch circuits **1301-1** to **1301-4** operate in accordance with the clock signal from the signal line CLK. The image data from the signal lines PDATA**4** to PDATA**1** and a second latch signal from a signal line LAT**2** are input to the latch circuits **1301-1** to **1301-4**. Upon receiving the second latch signal from the signal line LAT**2**, the latch circuits **1301-1** to **1301-4** latch the values being output to the signal lines PDATA**4** to PDATA**1** at that time, and output drive signals based on the latched values to the signal lines PON**2-4** to PON**2-1**, as illustrated in FIG. **14**. That is, the latch circuits **1301-1** to **1301-4** store the input image data, and output drive signals based on the stored image data. In other words, the latch circuits **1301-1** to **1301-4** correspond to a second data storage unit, or a plurality of data storage units. Furthermore, the second latch signal corresponds to a second trigger signal. The drive signals each output to the signal lines PON**2-4** to PON**2-1** are drive signals of the light emitting points **602** located next (right side in FIG. **5**) in the main scan direction to the light emitting point **602** driven by the drive signals output to the signal lines PON**1-4** to PON**1-1**. In other words, the drive signals output to the signal lines PON**2-4** to PON**2-1** are drive signals of each second light emitting point **602** of the four sets. According to FIG. **14**, the signal lines PON**2-1** to PON**2-4** output drive signals based on the pieces of image data D**5** to D**8**. The drive signals are input to the current drive unit **1104**. The latch units **1004-002** maintains outputting the drive signals based on the pieces of image data D**5** to D**8** until receiving a second latch signal based on the next line synchronization signal. In other words, the drive signal based on the pieces of image data D**5** to D**8** are output for the duration of two consecutive line synchronization signals (about 105.8 us in the present example). In addition, the latch unit **1004-002** delays the second latch signal by the D flip-flops **1302-1** to **1302-4**, and outputs a third latch signal to a signal line LAT**3** at a timing when the transfer unit **1003** outputs the piece of image data D**12** to the signal line PDATA**4**.

[0048] The same goes for the operation of the latch units **1004-003** to **1004-748**. In summary, each of the latch units **1004-***m* (m being an integer from 1 to 748) includes four latch circuits. The four latch circuits of the latch unit **1004-***m* each latch the pieces of image data D(4m), D(4m−1), D(4m−2), and D**4**(*m*−3) based on the m-th latch signal, and output drive signals based on the latched image data to the current drive unit **1104** for the duration of two consecutive line synchronization signals. Here, the four latch circuits of the latch unit **1004-***m* correspond to the m-th light emitting point **602** among the 748 light emitting points **602** arranged along the main scan direction of each of the first to the fourth sets. In other words, the four latch circuits of the latch unit **1004-***m* output the drive signals of the m-th light emitting point **602** of each of the first to the fourth sets. In addition, the latch units **1004-***m* except for m=748 generate an (m+1)-th latch signal based on the m-th latch signal and output the (m+1)-th latch signal to the latch unit **1004-**(*m*+1). As such, each of the latch units **1004-001** to **1004-748** outputs a drive signal for controlling emission/non-emission of light of each light emitting point **602** to the current drive unit **1104** for the duration of two consecutive line synchronization signals. In other words, the transfer unit **1003** and the latch units **1004-001** to **1004-748** correspond to a signal output unit. In addition, the latch circuits **1101-5** to **1101-8** included in the transfer unit **1003** and the latch circuits **1202-1** to **1202-4** included in the latch unit **1004** correspond to a signal output circuit.

[0049] FIG. **17** illustrates a relation between the line synchronization signal of the signal line SYNC, the first latch signal and the second latch signal of the signal lines LAT**1** and LAT**2**, and the drive signals output to the signal lines PON**1-1** to PON**1-4** and PON**2-1** to PON**2-4**. The cycle of the line synchronization signal is about 105.8 us as described above. The cycles of the first latch signal and the second latch signal are about 105.8 μs, which is same as that of the line synchronization signal. The timing of the second latch signal is behind the first latch signal by four clocks. More generally, the timing of the (q+1)-th latch signal (q being an integer from 1 to 747) is behind the q-th latch signal by four clocks.

[0050] The transfer unit **1003** sequentially receives the pieces of image data D**1**[**1**] to D**2992**[**1**] in synchronization with an initial line synchronization signal illustrated in FIG. **17**. In addition, the transfer unit **1003** sequentially receives the pieces of image data D**1**[**2**] to D**2992**[**2**] in synchronization with a second line synchronization signal illustrated in FIG. **17**. The four latch circuits of the latch unit **1004-001** output drive signals, based on the pieces of image data D**1**[**1**] to D**4**[**1**], to the signal lines PON**1-1** to PON**1-4** from the timing of the first latch signal based on the initial line synchronization signal illustrated in FIG. **17**. Output of drive signals based on the pieces of image data D**1**[**1**] to D**4**[**1**] continues until a first latch signal based on the second line synchronization signal is received. Upon receiving the first latch signal based on the second line synchronization signal, the four latch circuits of the latch unit **1004-001** output drive signals based on the pieces of image data D**1**[**2**] to D**4**[**2**] to the signal lines PON**1-1** to PON**1-4**. Similarly, the four latch circuits of the latch unit **1004-002** output drive signals based on the pieces of image data D**5**[**1**] to D**8**[**1**] to the signal lines PON**2-1** to PON**2-4** from the timing of the second latch signal based on the initial line synchronization signal. Output of drive signals based on the pieces of image data D**5**[**1**] to D**8**[**1**] continues until a second latch signal based on the second line synchronization signal is received. Upon receiving the second latch signal based on the second line synchronization signal, the four latch circuits of the latch unit **1004-002** output drive signals based on the pieces of image data D**5**[**2**] to D**8**[**2**] to the signal lines PON**2-1** to PON**2-4**. The same goes for the latch units **1004-003** to **1004-748**. In other words, each of the latch units **1004** are cascade-connected to each other such that they can sequentially transmit latch signals.

[0051] Here, the cycle of the line synchronization signal is longer than the time required for transmitting latch signals corresponding to all of the light emitting elements, and therefore all of the latch signals are output in a period from when a line synchronization signal is transmitted to when a next line synchronization signal is transmitted. Since the light emitting points **602** are sequentially driven in a state where the timing of starting the light emission is shifted for each of the four light emitting points, it is possible to reduce the noise generated at turning on the light emitting points, in comparison to the case of simultaneously turning on all the light emitting points. Furthermore, the output to the current drive unit **1104** is performed for the duration of two consecutive line synchronization signals.

[0052] In the present embodiment, the transfer unit **1003** and the latch unit **1004** form an output unit that receives image data corresponding to each of the plurality of light emitting points **602**, and outputs a drive signal for the corresponding light emitting point **602** based on the received image data. More specifically, each of the 748 latch units **1004-001** to **1004-748** of the present embodiment includes four latch circuits, and therefore the output unit includes a total of 2992 latch circuits. Each latch circuit corresponds to one of the 2992 light emitting points **602** included in one light emitting chip **400**. Each latch circuit latches image data of the corresponding light emitting point **602**, and outputs a drive signal for the corresponding light emitting point **602** based on the latched image data.

[0053] FIG. **15** illustrates a configuration of the current drive unit **1104**. Here, FIG. **15** illustrates only a part of the circuit corresponding to one light emitting point **602**. The light emitting chip **400** according to the present embodiment include a total of 2992 light emitting points **602**, and therefore the light emitting chip **400** include 2992 circuit units illustrated in FIG. **15**. A DAC **1501** outputs an analog voltage in accordance with a digital value indicated by the control data stored in the register **1102**. In an FET **1502**, which is a P-channel MOSFET, a source terminal is connected to a power supply voltage VCC and a drain terminal is connected to a source terminal of an FET **1503**. The analog voltage output by the DAC **1501** is applied to the gate terminal of the FET **1502**. Further, in the FET **1503**, which is also a P-channel MOSFET, a drain terminal is connected to the lower electrode **504**. The drive signal output from the image data holding unit **1103** is input to the gate terminal of the FET **1503** via a switching circuit **1504**. The drive signal is a one-bit binary signal of a high level or a low level, and the FET **1503** is on during the high level, and the FET **1503** is off during the low level.

[0054] While FET **1503** is on, electric current flows from the power supply voltage VCC to the light emitting layer **506** via FET **1502** and FET **1503** and thus the light emitting point **602** emits light. The light emission intensity of the light emitting point **602** varies in accordance with the current flowing through the light emitting layer **506**, and the value of the current is controlled by an analog voltage output from the DAC **1501**. In other words, the light emission intensity of each of the light emitting points **602** is controlled by the control data stored in the register **1102**. Here, the control data may individually indicate a digital value of each DAC **1501** corresponding to respective light emitting points **602**, or may indicate one digital value for each group of the plurality of light emitting points **602**. Here, the switching circuit **1504** is provided to switch between a normal state, in which the drive signal is applied to the gate terminal of the FET **1503**, and a test state. In the test state, the switching circuit **1504** applies a high level signal to the gate terminal of the FET **1503**, forcibly causing the FET **1503** to be on state. Here, the switching between the normal state and the test state is also performed based on control data stored in the register **1102**. The test state maybe used to cause any of the light emitting points **602** to emit light in manufacturing of the exposure head **106**.

[0055] FIG. **16** is a flowchart of processing executed by the image controller **700** when a print request is issued from a user. At S**10**, the image controller **700** writes, to the register **1102** of each of the light emitting chips **400**, control data for making the state of each switching circuit **1504** to be the normal state. At S**11**, the image controller **700** sets, in the register **1102** of each of the light emitting chips **400**, a digital value to be set in the DAC **1501** corresponding to each light emitting point **602**. Subsequently, at the start timing of image formation, the image controller **700** transmits an image data at S**12** and starts exposing the photoconductor **102**. The image controller **700** determines at S**13** whether or not the image formation is completed, and when the image formation is not completed, repeats the processing from S**12**. When, on the other hand, the image formation is completed, the image controller **700** terminates the processing of FIG. **13**.

[0056] Here, when each light emitting point **602** is made to emit light in manufacturing of the exposure head **106**, the image controller **700** first sets, in the register **1102**, a digital value to be set in the DAC **1501** corresponding to the target light emitting point **602**. Subsequently, the image controller **700** writes, to the register **1102**, control data for making the state of the switching circuit **1504** to be in the test state. After completion of the test, the image controller **700** writes control data for making the state of the switching circuit **1504** to be the normal state. Here, a configuration of making the light emitting points **602** to be the test state may be a configuration in which each light emitting point **602** is independently specified, or a configuration in which each light emitting point group including two or more light emitting points **602** is specified, for example, all the light emitting points **602** are collectively specified.

[0057] As has been described above, the latch unit **1004**-*m* in the present embodiment latches the pieces of image data D(4m), D(4m−1), D(4m−2) and D**4**(*m*−3) when the m-th latch signal is input, and outputs a drive signal based on the latched image data to the current drive unit **1104**. Such a configuration allows for turning on the plurality of light emitting units **602**, based on the image data transmitted in serial communication, without providing flip-flop circuits, configured to transfer the input image data in the main scan direction, in number corresponding to the number of light emitting units. In other words, increase in size of the light emitting chips or increase in cost of the light emitting chips due to the flip-flop circuits, configured to transfer the input image data in the main scan direction, being provided in number corresponding to the number of the light emitting units can be prevented. In other words, increase in size and in cost of the light emitting chips can be prevented more than the related art.

[0058] Generally, the amount of light emitted by organic EL film is smaller than the amount of light emitted by an LED formed by using gallium arsenide or the like, for example. In the present embodiment, drive signals for controlling emission/non-emission of light of each light emitting point **602** are output to the current drive unit **1104** for the duration of two consecutive line synchronization signals. That is, in the present embodiment, the light emitting point **602** may emit light for the duration of two consecutive line synchronization signals. As a result, exposing time for the photoconductor **102** can be made longer than a configuration in which the light emitting point **602** emits light for only a part of the duration of two consecutive line synchronization signals. As a result, the photoconductor **102** can be sufficiently exposed to form a toner image, whereby the toner image is appropriately formed on the photoconductor **102**. In other words, it is possible to prevent a case in which the photoconductor **102** is not sufficiently exposed to form a toner image due to the light emitting point **602** only emitting light for a part of the duration of two consecutive line synchronization signals.

[0059] When the FET **1503** is made to be on, there is a possibility that the power line voltage may fluctuate due to the light emitting point **602** being supplied with the current output from the power supply VCC, and thus radiation noise may be generated from the power line. In the present embodiment, the light emitting points **602** are sequentially turned on four by four, instead of simultaneously turning on all the light emitting points **602** provided in the light emitting chips **400**. As a result, voltage fluctuation of the power line is prevented in comparison with the case of simultaneously turning on the FET **1503** corresponding to each of the light emitting points **602** in order to simultaneously turn on all the light emitting points **602** provided in the light emitting chips **400**. As a result, the radiation noise generated from the power line can be reduced.

[0060] In addition, the light emission periods of the light emitting points **602** adjacent to each other in the main scan direction overlap with each other, in the duration of two consecutive line synchronization signals. When the light emission start timings of the light emitting points **602** are largely different, exposure positions of the light emitting points **602** in the sub scan direction may be largely different. By overlapping the exposure periods of the light emitting points **602** with each other, exposure time of the photoconductor **102** can be secured and the risk of misalignment of exposure positions in the sub scan direction for each light emitting point can be reduced.

[0061] Here, in the present embodiment, one light emitting chip **400** includes four sets of 748 light emitting points **602** arranged along the main scan direction. Accordingly, a total of 2992 latch circuits corresponding to each light emitting point **602** are grouped by each four latch circuits, which number corresponds to the number of sets, into a total of 748 groups, and one latch unit **1004** is provided for each corresponding group. In addition, since one group includes four latch circuits, the transfer unit **1003** sequentially transfers the image data to the four signal lines PDATA**1** to PDATA**4**. However, for example, the number of latch circuits included in one group may be an integer multiple of the number of sets. For example, when the number of latch circuits included in one group is set to be twice the number of sets for the light emitting chips **400** including four sets of 748 light emitting points **602** arranged along the main scan direction, the number of groups is 374, and the number of latch units **1004** is 374. Additionally in this case, the transfer unit **1003** sequentially transfers the image data to eight signal lines. In the present invention, the number of latch circuits included in one group is not limited to an integer multiple of the number of sets, and may be, for example, one over an integer of the number of sets. In addition, the number of latch circuits included in one group may be set irrespective of the number of sets. Furthermore, the number of light emitting points **602** (or the number of latch circuits) included in each group may not be the same. In other words, the number of light emitting points **602** (or the numbers of latch circuits) included in each group may be any number equal to or larger than one.

[0062] Although specific numerical values are used for description in the aforementioned embodiment, the specific numerical values are merely exemplary and the present invention is not limited to any specific numerical values used in the embodiment. Specifically, the number of the light emitting chips **400** provided on one printed substrate **202** is not limited to 20, and may be any number equal to or larger than one. In addition, the number of light emitting points **602** included in each of the light emitting chips **400** is not limited to 2992 and may be other numbers. Further, in the present embodiment, one light emitting chip **400** includes four sets of 748 light emitting points arranged along the main scan direction, the number of sets may be any number equal to or larger than one. In addition, the light emitting points **602** are arranged in the main scan direction at a pitch of about 21.16 μm, which corresponds to the resolution of 1200 dpi, the arrangement interval of the light emitting points **602** may be other values.

[0063] Further, in the aforementioned embodiment, the image forming apparatus transfers the toner image formed on each photoconductor **102** onto the sheet conveyed on the transfer belt **111**.

However, the image forming apparatus may transfer the toner image of each photoconductor **102** onto the sheet via an intermediate transfer member. In addition, the image forming apparatus may be a color image forming apparatus that forms an image using toner of a plurality of colors, or a monochrome image forming apparatus that forms an image using toner of one color.

OTHER EMBODIMENTS

[0064] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0065] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0066] This application claims the benefit of Japanese Patent Application No. 2022-132718, filed Aug. 23, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A light emitting chip comprising: a light emitting unit array including a plurality of light emitting units arranged in a predetermined direction; an image data input terminal configured to be input an image data; a first data storage unit configured to store data, input from the image data input terminal, for causing a first light emitting unit included in the plurality of light emitting units to emit light; a second data storage unit configured to store data, input from the image data input terminal, for causing a second light emitting unit included in the plurality of light emitting units to emit light; a signal output unit configured to output a first trigger signal and a second trigger signal delayed with respect to the first trigger signal; and a drive unit configured to drive the first light emitting unit, in a case where the first trigger signal is output, based on data stored in the first data storage unit, and drive the second light emitting unit, in a case where the second trigger signal is output, based on data stored in the second data storage unit.

2-18. (canceled)